

DD: PLM-CFG-01 v1.0 — Service Catalog

Developer implementation guide: DD_PLM-CFG-01-Service_Catalog-DEV_IMPL_GUIDE-v1.0.md.

1. Purpose

This rewritten DD is the developer-facing version of the module contract DD_PLM-CFG-01-Service_Catalog-v1.0.md.

Verdict: ■ Updated | **Wave:** 1 | **Group I:** Product Catalog | **Version:** v1.0 The Service Catalog. Owns: the `service_class` table (admin-managed classes: Internet, Voice, IPTV, Monitoring, ...), the `service` table (the leaf Services that Packages compose), the Provisioner adapter interface declaration, and the rules governing addressable vs non-addressable Services. **What changed from v0.8:** - **Added** `service_class.requires_equipment` boolean — declares whether a service class needs customer-premises equipment to fulfill. Used by UPGRADE-01 / DOWNGRADE-01 / MIGRATION-01 for equipment-delta computation when comparing source vs target Package compositions. - **Added** `service_class.service_management_key_default` — the canonical HomePass endpoint key per class (e.g., Internet class → `data`, IPTV class → `iptv_multicast`). Services inherit this default; per-Service override remains. - **Formal Provisioner interface declaration** — 5 methods (`activateService` / `modifyService` / `deactivateService` / `resumeService` / `suspendService`). Referenced by FUL-03 v1.1 (first four) + FUL-04 v1.0 (`suspendService` + restriction verbs not on this interface). - **Service version pinning contract** — Services no longer carry rate-cards or pricing in this catalog; pricing lives in BIL-CFG-01 BillableEvents (for ONE_SHOT) and SIP-01 Package Versions (for CONTINUOUS). PLM-CFG-01 v1.0 strictly owns the Service shape (class, addressability, equipment-requirement, network-profile-shape, service-management-key, provisioner-key) — not its price.

The goal of this rewrite is to make the DD easier for a mid-level developer to implement without losing the detailed source contract.

2. Implementation Scope

This DD should be implemented as part of the owning SOPHIX capability, not as an isolated service unless the deployment architecture explicitly separates that capability.

Implement this DD with these rules:

- Use the owning module APIs/repositories for authoritative writes.
- Use approved internal APIs/client wrappers when reading another module's data.
- Do not query another module's database tables directly from business flow code.
- Treat worker names as logical Camunda external-task topics, not separate deployments.
- One worker application may handle multiple topics, but metrics, retries, and logs must remain visible per topic.
- Emit success events only after the authoritative state commit succeeds.
- Use outbox publishing for Kafka events.

3. Runtime Metadata

Item	Value
Source file	/Users/macbook/Downloads/Sophix_V3_BSS_DDs_MD_2026-05-27/09_catalogs/DD_PLM-CFG-01-Service_Catalog-v1.0.md
Version	v1.0
DD type	module contract
BPMN/process keys	Not explicitly defined
Drools packages	Not explicitly defined

4. Runtime Contracts To Implement

4.1 APIs

- DELETE `/api/plm/service-classes/{id}`
- DELETE `/api/plm/services/{id}`

- GET /api/plm/service-classes/{id}
- GET /api/plm/service-classes?operatorCode=WIK&retired=false
- GET /api/plm/services/{id}
- GET /api/plm/services?operatorCode=WIK&serviceClassId=scl_...&consumptionModel=CONTINUOUS
- PATCH /api/plm/service-classes/{id}
- PATCH /api/plm/services/{id}
- POST /api/plm/service-classes
- POST /api/plm/services

4.2 Tables

- service
- service_class

For each table in this DD, implement the schema with:

- a short table comment explaining what the table is for;
- column comments or migration notes explaining each field;
- constraints and indexes from the DD;
- seed data where the DD provides operator-specific sample rows.

4.3 Worker Topics

- gpon-res
- hfc-docsis
- verimatrix-iptv
- voip-sip

Worker-topic guidance:

- A BPMN service task uses the topic.
- A worker application subscribes to the topic.
- The topic handler validates input variables, calls the owning module/API, writes outputs back to Camunda, and records metrics.
- Do not treat the topic string as a shell command or Kubernetes deployment name.

4.4 Events

- No domain events are explicitly declared in this DD.

Event guidance:

- Write events through the transactional outbox.
- Include operation id, aggregate id, operator code, actor, and correlation data.
- Consumers should be able to rebuild downstream state without querying workflow internals.

5. Developer Implementation Checklist

1. Add or verify database migrations and seed rows.
2. Add or verify config rows for the operator/module.
3. Implement request/command DTOs and validation.
4. Implement idempotency and in-flight operation checks where the DD requires them.
5. Implement Drools fact mapping and package deployment where rules are used.
6. Implement BPMN process, service tasks, gateways, timers, and message catches where the DD is a flow.

7. Implement worker-topic handlers using owning module APIs.
8. Implement compensation paths and manual recovery paths.
9. Emit success/rejected events through the outbox.
10. Add smoke tests for happy path, validation failure, dependency failure, and retry/idempotency.

6. Infrastructure And AWS Notes

Deploy this DD with the existing SOPHIX platform components unless the owning module requires isolation:

Concern	Recommendation
API/runtime	Run inside the owning module deployment or existing workflow API pods.
Workers	Consolidate low-volume topics into shared worker apps; keep per-topic metrics.
Database	Use the owning module PostgreSQL schema and migration pipeline.
Kafka	Use the foundation Kafka/MSK setup and transactional outbox.
Camunda	Deploy BPMN files through the workflow deployment pipeline.
Drools	Deploy KJAR/rule artifacts through the approved rules pipeline.
Secrets	Use the platform secret manager; on AWS prefer Secrets Manager/Parameter Store with IRSA.
Observability	Alert on stuck operations, failed workers, outbox backlog, and external dependency failures.

7. Original DD Detail Preserved

The following section preserves the detailed source contract for implementation and review.

Verdict: ■ Updated | **Wave:** 1 | **Group I:** Product Catalog | **Version:** v1.0

The Service Catalog. Owns: the `service_class` table (admin-managed classes: Internet, Voice, IPTV, Monitoring, ...), the `service` table (the leaf Services that Packages compose), the Provisioner adapter interface declaration, and the rules governing addressable vs non-addressable Services.

What changed from v0.8:

- **Added** `service_class.requires_equipment` boolean — declares whether a service class needs customer-premises equipment to fulfill. Used by UPGRADE-01 / DOWNGRADE-01 / MIGRATION-01 for equipment-delta computation when comparing source vs target Package compositions.
- **Added** `service_class.service_management_key_default` — the canonical HomePass endpoint key per class (e.g., Internet class → `data`, IPTV class → `iptv_multicast`). Services inherit this default; per-Service override remains.
- **Formal Provisioner interface declaration** — 5 methods (`activateService` / `modifyService` / `deactivateService` / `resumeService` / `suspendService`). Referenced by FUL-03 v1.1 (first four) + FUL-04 v1.0 (`suspendService` + restriction verbs not on this interface).
- **Service version pinning contract** — Services no longer carry rate-cards or pricing in this catalog; pricing lives in BIL-CFG-01 BillableEvents (for ONE_SHOT) and SIP-01 Package Versions (for CONTINUOUS). PLM-CFG-01 v1.0 strictly owns the Service shape (class, addressability, equipment-requirement, network-profile-shape, service-management-key, provisioner-key) — not its price.

Glossary

Term	Meaning
Service class	A coarse-grained category of fulfillment: Internet, Voice, IPTV, Monitoring, Mobile, etc. Per-operator catalog. Carries class-level defaults (tax group, wallet, equipment requirement, canonical service-management key).
Service	A specific leaf product within a class — e.g., "Internet 100Mbps GPON", "VoIP standard", "IPTV basic tier". Belongs to one class. Bundled into Packages (per SIP-01).

Term	Meaning
Addressable Service	A Service whose activation requires equipment + a network profile + a HomePass endpoint resolution. CONTINUOUS Services in Packages are typically addressable. ONE_SHOT Services (fees, ETFs) are typically non-addressable.
Consumption model	CONTINUOUS (recurring; accrues across bill cycles; belongs to Packages) or ONE_SHOT (single charge on a trigger; referenced by BillableEvents in BIL-CFG-01).
Provisioner key	The identifier of the adapter that translates a FUL-03 / FUL-04 / FUL-05 verb into vendor commands. Examples: gpon-res (Huawei NCE), hfc-docsis (Casa BSR), voip-sip (SIP switch), verimatrix-iptv (Verimatrix middleware).
Network profile shape	The set of parameters the Provisioner adapter needs to fulfill: speed tier, codec, multicast groups, etc. Carried as JSONB {provisionerKey, params} per Service.
Service management key	The key looked up in the HomePass's service_management_endpoints map (per RLM-CFG-01) to resolve {nodeCode, port}. Examples: data, voice, iptv_multicast, monitoring. Per-deployment configurable.
requires_equipment	Class-level flag declaring whether this service class needs customer-premises equipment (ONT, ATA, STB, modem) to fulfill. Drives UPGRADE/DOWNGRADE/MIGRATION's equipment-delta logic.

1. What this is

The Service Catalog. Two entities: `service_class` (the classes the operator sells — Internet, Voice, IPTV, ...) and `service` (the specific leaf products within a class — Internet 100Mbps GPON, VoIP standard, ...). Services are bundled into Packages by SIP-01.

Who reads from PLM-CFG-01:

- **SIP-01** — reads Service definitions when validating a Package's `serviceRefs[]`.
- **FUL-03 / FUL-04 / FUL-05** — read Service definitions to resolve `service_management_key` → HomePass endpoint, dispatch to Provisioner adapter via `provisioner_key`, apply `network_profile_shape.params`.
- **UPGRADE-01 / DOWNGRADE-01 / MIGRATION-01** — read `service_class.requires_equipment` to compute equipment-delta when comparing source vs target Package compositions.
- **BIL-01 / BIL-CFG-01** — read Service `consumption_model` to enforce that CONTINUOUS Services are referenced by Packages and ONE_SHOT Services by BillableEvents.
- **CVM / Reporting** — read `revenue_category` for finance/analytics revenue stream separation.

What this DD owns:

1. The `service_class` and `service` tables.
2. The Provisioner adapter interface declaration (5 methods).
3. The rules governing addressability + consumption model + equipment requirement.
4. CRUD APIs for admin-managed catalog entries.
5. Kafka events on catalog changes (consumed by SIP-01 + downstream).

What this DD does NOT own:

- **Package composition** — owned by SIP-01.
- **Service pricing** — pricing lives in BIL-CFG-01 (BillableEvents for ONE_SHOT) and SIP-01 Package Versions (for CONTINUOUS Package prices). PLM-CFG-01 carries default tax group and default wallet refs for revenue allocation, but never a price.
- **Provisioner adapter implementations** — those are code beans in the fulfillment module, registered per `provisioner_key`. PLM-CFG-01 declares the interface contract; the beans live in code.
- **HomePass service_management_endpoints map** — owned by RLM-CFG-01. PLM-CFG-01 only declares which keys Services consult.
- **Equipment binding to subscriptions** — owned by OSR-INSTANCE-01 (the equipment inventory module). PLM-CFG-01 declares the requirement; OSR-INSTANCE-01 tracks the binding.

2. Schemas

2.1 service_class

Admin-managed list of service classes. One row per (operator, class).

```
CREATE TABLE service_class (  
  id TEXT PRIMARY KEY,  
  operator_code TEXT NOT NULL,  
  name TEXT NOT NULL,  
  description TEXT NULL,  
  requires_equipment BOOLEAN NOT NULL DEFAULT FALSE,  
  service_management_key_default TEXT NULL,  
  default_tax_group_ref TEXT NOT NULL,  
  default_wallet_ref TEXT NOT NULL,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
  created_by TEXT NOT NULL,  
  updated_at TIMESTAMPTZ NULL,  
  updated_by TEXT NULL,  
  retired_at TIMESTAMPTZ NULL,  
  UNIQUE (operator_code, name) WHERE retired_at IS NULL  
);
```

Column	Notes
id	ULID. Example: scl_01HX_INET
operator_code	WIK / WUG / WTZ / YASSN. Classes are operator-scoped — each operator can declare its own taxonomy.
name	Display name, unique per operator. Examples: Internet, Voice, IPTV, Monitoring, Mobile.
requires_equipment	true when this class needs customer-premises equipment (ONT for Internet, ATA for Voice, STB for IPTV). false for software/cloud services (Mobile data plan, OTT bundle). Drives UPGRADE/DOWNGRADE/MIGRATION's equipment-delta logic — when target Package adds a class with requires_equipment=true and the customer doesn't have suitable equipment bound, the workflow forces equipmentPath=NEW_INSTALL.
service_management_key_default	The canonical HomePass endpoint key for this class. NULL for non-addressable classes (cloud-only). Examples: Internet → data, Voice → voice, IPTV → iptv_multicast, Monitoring → monitoring. Services inherit this default; per-Service override remains via the Service's service_management_key column.
default_tax_group_ref	String ref to PLM-CFG-02 tax group. Used as default for Services in this class; per-Service override allowed.
default_wallet_ref	String ref to PLM-CFG-03 wallet. Used as default for Services in this class.

Sample data:

id	operator_code	name	requires_equipment	service_management_key_default	default_tax_group_ref	default_wallet_ref
scl_01HX_INET_WIK	WIK	Internet	true	data	tg_VAT_16_WIK	wlt_WIK_AIRTIME
scl_01HX_VOIP_WIK	WIK	Voice	true	voice	tg_VAT_16_WIK	wlt_WIK_AIRTIME
scl_01HX_IPTV_WIK	WIK	IPTV	true	iptv_multicast	tg_VAT_16_WIK	wlt_WIK_AIRTIME
scl_01HX_MON_WIK	WIK	Monitoring	false	monitoring	tg_VAT_16_WIK	wlt_WIK_AIRTIME
scl_01HX_OTT_WIK	WIK	OTT_Bundle	false	NULL	tg_VAT_16_WIK	wlt_WIK_AIRTIME

(Row 5: an OTT bundle — Netflix-style add-on — is non-addressable, no HomePass endpoint needed, no equipment.)

2.2 service

The Service catalog. One row per (operator, service code).

```
CREATE TABLE service (  
  id TEXT PRIMARY KEY,  
  operator_code TEXT NOT NULL,
```

```

code TEXT NOT NULL,
description TEXT NULL,
service_class_id TEXT NOT NULL REFERENCES service_class(id),
service_group TEXT NULL,
is_addressable BOOLEAN NOT NULL DEFAULT FALSE,
consumption_model TEXT NOT NULL,
revenue_category TEXT NULL,
equipment_requirement_ref TEXT NULL,
network_profile_shape JSONB NULL,
provisioner_key TEXT NULL,
service_management_key TEXT NULL,
default_wallet_ref TEXT NOT NULL,
default_tax_group_ref TEXT NOT NULL,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by TEXT NOT NULL,
updated_at TIMESTAMPTZ NULL,
updated_by TEXT NULL,
retired_at TIMESTAMPTZ NULL,
UNIQUE (operator_code, code) WHERE retired_at IS NULL,
CHECK (consumption_model IN ('CONTINUOUS', 'ONE_SHOT')),
CHECK (NOT is_addressable OR (provisioner_key IS NOT NULL AND network_profile_shape IS NOT NULL))
);

```

Column	Notes
id	ULID. Example: svc_01HX_INET_100M_GPON_WIK
code	Unique business code per operator. Example: INET_100M_GPON.
service_class_id	FK to service_class.
is_addressable	true when activation needs equipment + network profile + HomePass endpoint resolution. CONTINUOUS Services in Packages are typically addressable; ONE_SHOT fee Services are typically non-addressable. Check constraint enforces that addressable Services have provisioner_key + network_profile_shape set.
consumption_model	CONTINUOUS (recurring; in Packages; e.g., Internet 100Mbps) or ONE_SHOT (single charge; referenced by BillableEvents; e.g., installation fee, ETF). Immutable after create.
revenue_category	Optional finance classification mirroring BIL-CFG-01's billable_event_category (RECURRING_INTERNET, RECURRING_VOICE, INSTALLATION_FEE, ETF, VAS, REGULATORY, RECURRING_IPTV, ...). Used by finance/analytics for revenue stream separation.
equipment_requirement_ref	String ref to PLM-CFG-06 equipment-type binding. Required when is_addressable=true AND class has requires_equipment=true. Allows per-Service equipment-type override (e.g., GPON 1Gbps Service might require a newer ONT model than the 100Mbps Service in the same class).
network_profile_shape	JSONB { "params": { ... } }. Examples: Internet 100Mbps → {"speedDownMbps": 100, "speedUpMbps": 20}; IPTV Basic → {"multicastGroups": ["epg_basic"], "stbAaaProfile": "basic-tier"}. Required when is_addressable=true.
provisioner_key	The Provisioner adapter that handles this Service. Required when is_addressable=true. Examples: gpon-res, hfc-docsis, voip-sip, verimatrix-iptv.
service_management_key	The key looked up in the HomePass's service_management_endpoints map. NULL inherits the class default. Required (or class default) when is_addressable=true.

Sample data:

id	operator_code	code	service_class_id	is_addressable	consumption_model	provisioner_key	service_management_key	revenue_category
svc_01HX_INET_100M_GPON_WIK	WIK	INET_100M_GPON	sc1_01HX_INET_WIK	true	CONTINUOUS	gpon-res	data	RECURRING_INTERNET
svc_01HX_VOIP_STD_WIK	WIK	VOIP_STD	sc1_01HX_VOIP_WIK	true	CONTINUOUS	voip-sip	voice	RECURRING_VOICE
svc_01HX_IPTV_BASIC_WIK	WIK	IPTV_BASIC	sc1_01HX_IPTV_WIK	true	CONTINUOUS	verimatrix-iptv	iptv_multicast	RECURRING_IPTV
svc_01HX_INSTALL_FEE_WIK	WIK	INSTALL_FEE	sc1_01HX_INET_WIK	false	ONE_SHOT	NULL	NULL	INSTALLATION_FEE
svc_01HX_ETF_WIK	WIK	EARLY_TERMINATION_FEE	sc1_01HX_INET_WIK	false	ONE_SHOT	NULL	NULL	ETF

(Rows 4-5: ONE_SHOT non-addressable fee Services. Referenced by BIL-CFG-01 BillableEvents, not by Packages.)

network_profile_shape JSONB samples:

```
// svc_01HX_INET_100M_GPON_WIK
{"params": {"speedDownMbps": 100, "speedUpMbps": 20, "burstAllowanceGB": 10}}

// svc_01HX_VOIP_STD_WIK
{"params": {"codec": "G.711a", "didCount": 1, "callForwardingEnabled": true}}

// svc_01HX_IPTV_BASIC_WIK
{"params": {"multicastGroups": ["epg_zuku_basic"], "stbAaaProfile": "basic-tier", "concurrentStreamLimit": 2}}
```

3. Provisioner adapter interface

Declared formally in PLM-CFG-01. Implementation beans live in the fulfillment module, registered per provisioner_key. All beans implement the same interface so FUL-03 / FUL-04 / FUL-05 can dispatch generically.

3.1 Interface declaration

```
public interface ServiceProvisioner {
    /** Stand up a Service at the customer's premises for the first time. Called by FUL-03 (ACTIVATE_SERVICE, MODIFY_SERVICE).
     * ProvisionerOutcome activateService(
     *     EquipmentBinding equipment,
     *     Map<String, Object> params,
     *     Customer customer,
     *     String targetNodeCode,
     *     String targetPort
     * );

    /** Change a running Service's parameters in place (e.g., speed tier change). Called by FUL-03 (MODIFY_SERVICE-modified).
     * ProvisionerOutcome modifyService(
     *     Map<String, Object> oldParams,
     *     Map<String, Object> newParams,
     *     Customer customer,
     *     String targetNodeCode,
     *     String targetPort
     * );

    /** Tear down a running Service permanently (post-termination network revocation). Called by FUL-03 (MODIFY_SERVICE-drop).
     * ProvisionerOutcome deactivateService(
     *     Map<String, Object> params,
     *     Customer customer,
     *     String targetNodeCode,
     *     String targetPort
     * );

    /** Temporarily suspend a running Service (customer-initiated PAUSE, non-payment SUSPEND_NP, or workflow compensation).
     * ProvisionerOutcome suspendService(
     *     String reasonCategory,
     *     Map<String, Object> params,
     *     Customer customer,
     *     String targetNodeCode,
     *     String targetPort
     * );

    /** Restore a suspended Service. Called by FUL-03 (RESUME_SERVICE, post-PAUSE). Also used as FUL-04's compensation inverse.
     * ProvisionerOutcome resumeService(
     *     Map<String, Object> params,
     *     Customer customer,
     *     String targetNodeCode,
     *     String targetPort
     * );
}
```

3.2 ProvisionerOutcome contract

```
public class ProvisionerOutcome {
    boolean success;
    Map<String, String> adapterRefs; // vendor-system IDs returned by the call (e.g., {"huaweiCommandId": "cmd_123"})
    String failureReason; // populated when success=false; coded enum (ADAPTER_TIMEOUT, ADAPTER_ERROR, UNREACHABLE)
    String failureDetail; // free-text human-readable
    long durationMs;
}
```

3.3 Adapter registration

Adapters register a bean in the fulfillment module's Spring context with `@ServiceProvisioner(key = "gpon-res")` (or `hfc-docsis`, `voip-sip`, etc.). FUL-03 / FUL-04 / FUL-05 look up the bean by `provisioner_key` from the Service's `provisioner_key` column.

A Service whose `provisioner_key` references an unregistered key fails at dispatch time with `UNKNOWN_PROVISIONER_KEY`.

3.4 Provisioner restrictions (FUL-04-only methods)

FUL-04 (Restriction Fulfillment) requires two additional methods on the adapter: `applyRestriction` and `liftRestriction`. These are **not on the ServiceProvisioner interface** — they're on a separate `RestrictionCapableProvisioner` extension interface. Adapters that support restrictions (e.g., AAA-capable adapters) implement both; adapters that don't (e.g., partner-fulfilled OTT bundle adapters) implement only `ServiceProvisioner`.

```
public interface RestrictionCapableProvisioner extends ServiceProvisioner {
    ProvisionerOutcome applyRestriction(String fulfillmentAction, String restrictionCode, Map<String,Object> params, Customer
    ProvisionerOutcome liftRestriction(String fulfillmentAction, String restrictionCode, Map<String,Object> params, Customer
}
```

FUL-04 checks the adapter implements `RestrictionCapableProvisioner` before invoking restriction methods; if not, returns `RESTRICTION_NOT_SUPPORTED_BY_PROVISIONER` for that Service.

4. Rules

4.1 Service class

Rule	Description
R-PLM-CFG-01-CL-1	A <code>service_class</code> row is operator-scoped. Operators may declare different class taxonomies (e.g., WIK may have IPTV class; WUG may not yet).
R-PLM-CFG-01-CL-2	<code>name</code> is unique per operator (excluding retired). Two operators may both have a class named Internet; this is the typical case.
R-PLM-CFG-01-CL-3	<code>requires_equipment</code> is set at create time per operator policy. Changing it post-create is allowed but requires manual ops review (audit log entry mandatory) — changing it retroactively affects UPGRADE/DOWNGRADE/MIGRATION decisions for in-flight workflows.
R-PLM-CFG-01-CL-4	<code>service_management_key_default</code> may be NULL (for non-addressable classes like OTT bundles). When non-null, Services in this class inherit the value unless the Service's own <code>service_management_key</code> overrides.
R-PLM-CFG-01-CL-5	A service class with active Services (Services not retired and referenced by a non-retired Package) cannot itself be retired. Returns 409 <code>SERVICE_CLASS_IN_USE</code> .

4.2 Service

Rule	Description
R-PLM-CFG-01-SV-1	A service row is operator-scoped. <code>code</code> is unique per operator (excluding retired).
R-PLM-CFG-01-SV-2	<code>service_class_id</code> MUST reference a non-retired <code>service_class</code> row in the same operator. Cross-operator references return 400 <code>SERVICE_CLASS_OPERATOR_MISMATCH</code> .
R-PLM-CFG-01-SV-3	<code>consumption_model</code> is immutable after create. Changing requires retire + re-create.
R-PLM-CFG-01-SV-4	Addressable Services (<code>is_addressable=true</code>) MUST have <code>provisioner_key</code> , <code>network_profile_shape</code> , and <code>service_management_key</code> set (or inherited from class). Check constraint enforced at insert.
R-PLM-CFG-01-SV-5	<code>equipment_requirement_ref</code> is required when <code>is_addressable=true</code> AND the parent service class has <code>requires_equipment=true</code> . Returns 400 <code>EQUIPMENT_REQUIREMENT_MISSING</code> otherwise.
R-PLM-CFG-01-SV-6	<code>service_management_key</code> inherits from <code>service_class.service_management_key_default</code> when NULL on the Service. The effective key at dispatch time is <code>service.service_management_key</code> ?? <code>service_class.service_management_key_default</code> . NULL on both for addressable Services is a configuration error → 400 at create.

Rule	Description
R-PLM-CFG-01-SV-7	Retiring a Service requires no active Packages reference it (per SIP-01 cross-DD check). Returns 409 SERVICE_IN_USE otherwise.

4.3 Provisioner

Rule	Description
R-PLM-CFG-01-PR-1	provisioner_key is a free-text string at the catalog level. Validation that a registered adapter exists is deferred to dispatch time (FUL-03 / FUL-04 / FUL-05). A Service may be created with an unregistered provisioner_key (e.g., pre-deployment); attempts to activate will fail at runtime with UNKNOWN_PROVISIONER_KEY.
R-PLM-CFG-01-PR-2	The 5-method ServiceProvisioner interface (§3.1) is the canonical contract. All adapters MUST implement all 5 methods (may return success=false, failureReason=NOT_SUPPORTED for methods they cannot perform — e.g., a partner-fulfilled adapter may not support modifyService).
R-PLM-CFG-01-PR-3	The RestrictionCapableProvisioner interface (§3.4) is optional. Adapters implement it only when they support restrictions.

4.4 Equipment requirement

Rule	Description
R-PLM-CFG-01-EQ-1	service_class.requires_equipment is the authoritative class-level flag. Used by UPGRADE-01 / DOWNGRADE-01 / MIGRATION-01 when computing equipment-delta between source and target Package compositions.
R-PLM-CFG-01-EQ-2	Per-Service equipment_requirement_ref declares the SPECIFIC equipment type. The class-level requires_equipment is a quick filter ("does this class need ANY equipment?"); the Service-level ref tells the workflow which equipment-type to look for in existing bindings or request from inventory.
R-PLM-CFG-01-EQ-3	Multi-Service equipment retention: when UPGRADE/DOWNGRADE/MIGRATION drops a class, the customer's existing equipment is retained if it ALSO fulfills another retained class (e.g., ATA fulfilling both VOIP and a future fax service; if VOIP class is dropped but fax class is retained, ATA stays). The workflow checks equipment_requirement_ref per class to determine multi-service overlap.

5. REST API

5.1 Service class CRUD

POST /api/plm/service-classes HTTP/1.1
Content-Type: application/json

```
{
  "operatorCode":      "WIK",
  "name":              "IPTV",
  "requiresEquipment": true,
  "serviceManagementKeyDefault": "iptv_multicast",
  "defaultTaxGroupRef": "tg_VAT_16_WIK",
  "defaultWalletRef":   "wlt_WIK_AIRTIME",
  "createdBy":         "admin-wik-001"
}
```

→ 201 Created

```
{
  "id":                "scl_01HX_IPTV_WIK",
  "operatorCode":      "WIK",
  "name":              "IPTV",
  "requiresEquipment": true,
  "serviceManagementKeyDefault": "iptv_multicast",
  "createdAt":         "2026-05-25T14:00:00+03:00"
}
```

GET /api/plm/service-classes/{id} — read. GET
/api/plm/service-classes?operatorCode=WIK&retired=false — list. PATCH
/api/plm/service-classes/{id} — update (audit logs changes). DELETE /api/plm/service-classes/{id} — soft-delete (sets retired_at).

5.2 Service CRUD

POST /api/plm/services HTTP/1.1
Content-Type: application/json

```
{
  "operatorCode":      "WIK",
  "code":              "IPTV_BASIC",
  "description":       "IPTV basic tier - 50 channels",
  "serviceClassId":    "scl_01HX_IPTV_WIK",
  "isAddressable":     true,
  "consumptionModel":  "CONTINUOUS",
  "revenueCategory":   "RECURRING_IPTV",
  "equipmentRequirementRef": "eqt_STB_BASIC",
  "networkProfileShape": {"params": {"multicastGroups": ["epg_zuku_basic"], "stbAaaProfile": "basic-tier", "concurrentStreams": 1}},
  "provisionerKey":     "verimatrix-iptv",
  "serviceManagementKey": null,
  "defaultWalletRef":   "wlt_WIK_AIRTIME",
  "defaultTaxGroupRef": "tg_VAT_16_WIK",
  "createdBy":          "admin-wik-001"
}

→ 201 Created
{
  "id":                "svc_01HX_IPTV_BASIC_WIK",
  "operatorCode":      "WIK",
  "code":              "IPTV_BASIC",
  "serviceClassId":    "scl_01HX_IPTV_WIK",
  "effectiveServiceManagementKey": "iptv_multicast",
  "createdAt":         "2026-05-25T14:05:00+03:00"
}
```

(effectiveServiceManagementKey is derived from the class default since the Service's own field is NULL.)

GET /api/plm/services/{id} — read. GET

/api/plm/services?operatorCode=WIK&serviceClassId=scl_...&consumptionModel=CONTINUOUS — filter.

PATCH /api/plm/services/{id} — update (audit logs changes; rejects changes to consumption_model). DELETE

/api/plm/services/{id} — soft-delete (R-SV-7 check).

5.3 Errors

```
→ 400 Bad Request
{
  "code":      "EQUIPMENT_REQUIREMENT_MISSING",
  "guidance":  "Service is addressable AND parent class has requires_equipment=true, but equipment_requirement_ref is null."
}

→ 409 Conflict
{
  "code":                "SERVICE_IN_USE",
  "referencingPackages": ["pkg_01HX_INET_VOIP_IPTV_RES"],
  "guidance":            "Service is referenced by 1 active Package. Retire the Package first, or replace the Service in the Package."
}
```

6. Kafka events

Event	Topic	Fires on	Consumers
ServiceClassCreated	sophix.plm.service-class.created	New service_class row	SIP-01, CVM, EM-04
ServiceClassUpdated	sophix.plm.service-class.updated	Mutable column change	SIP-01, CVM
ServiceClassRetired	sophix.plm.service-class.retired	retired_at set	SIP-01, CVM
ServiceCreated	sophix.plm.service.created	New service row	SIP-01, BIL-CFG-01, FUL-03 cache invalidation
ServiceUpdated	sophix.plm.service.updated	Mutable column change	SIP-01, FUL-03 cache invalidation
ServiceRetired	sophix.plm.service.retired	retired_at set	SIP-01, CVM

Sample ServiceCreated:

```
{
  "eventType":  "ServiceCreated",
  "aggregateId": "svc_01HX_IPTV_BASIC_WIK",
}
```

```

"timestamp": "2026-05-25T14:05:00+03:00",
"payload": {
  "operatorCode": "WIK",
  "id": "svc_01HX_IPTV_BASIC_WIK",
  "code": "IPTV_BASIC",
  "serviceClassId": "scl_01HX_IPTV_WIK",
  "isAddressable": true,
  "consumptionModel": "CONTINUOUS",
  "revenueCategory": "RECURRING_IPTV",
  "provisionerKey": "verimatrix-iptv",
  "effectiveServiceManagementKey": "iptv_multicast",
  "createdBy": "admin-wik-001"
}
}

```

7. Cross-DD coordination

Module	Direction	Contract
SIP-01	Consumer	Reads Service definitions when validating Package serviceRefs[]. Consumes catalog events for cache invalidation.
FUL-03 v1.1	Consumer	Reads Service service_management_key, provisioner_key, network_profile_shape, consumption_model, is_addressable, equipment_requirement_ref. Implements ServiceProvisioner interface adapters per §3.
FUL-04 v1.0	Consumer	Same fields as FUL-03. Implements RestrictionCapableProvisioner interface adapters per §3.4.
FUL-05 v1.0	Consumer	Reads Service definitions for termination revocation. Implements ServiceProvisioner.deactivateService per §3.
SUB-WF-UPGRADE-01 / DOWNGRADE-01 / MIGRATION-01	Consumer	Read service_class.requires_equipment for equipment-delta computation. Read service.equipment_requirement_ref for specific equipment-type lookup.
BIL-CFG-01	Consumer	Reads ONE_SHOT Services for BillableEvent definitions. Reads revenue_category for revenue stream classification.
PLM-CFG-02 (Tax)	Referenced	default_tax_group_ref on both service_class and service references PLM-CFG-02.
PLM-CFG-03 (Wallet)	Referenced	default_wallet_ref references PLM-CFG-03.
PLM-CFG-06 (Equipment Type Catalog)	Referenced	service.equipment_requirement_ref references PLM-CFG-06.

7.1 Build order

This DD (v1.0) deploys after:

1. PLM-CFG-02 v1.0 (tax groups)
2. PLM-CFG-03 v1.0 (wallets)
3. PLM-CFG-06 v1.0 (equipment type catalog)

Callers depending on v1.0:

- SIP-01 v1.0 (Package composition)
- FUL-03 v1.1 (5-method Provisioner interface)
- FUL-04 v1.0 (already deployed against v0.8 — backwards-compatible)
- FUL-05 v1.0 (already deployed against v0.8 — backwards-compatible)
- UPGRADE-01 / DOWNGRADE-01 / MIGRATION-01 (require service_class.requires_equipment)

7.2 Stub debt + open coordination items

- **v0.8 → v1.0 migration** — production `service_class` rows need `requires_equipment` + `service_management_key_default` backfilled. KE WIK seed: `Internet=true, Voice=true, IPTV=true, Monitoring=false, OTT_Bundle=false`. Migration script ships with v1.0.
- **Adapter registration audit** — once v1.0 ships, ops should audit existing `Service` rows for `provisioner_key` values that don't match registered adapters and either retire or update.
- **PLM-CFG-06 Equipment Type Catalog v1.0** — pending; provides equipment types referenced by `service.equipment_requirement_ref`.

B — Current TZ

Status: To be filled in after codebase cartography review.

The TZ codebase has a `Service` catalog implementation (the v0.8 baseline was authored against partial observation). Specific structural details — whether `service_class.requires_equipment` exists today, whether the 5-method `Provisioner` interface is formally declared in code, whether `service_management_key` defaults are class-level or per-`Service` — require source-code review to document accurately. Until cartography lands, this DD does not assert specific legacy structural claims.